Education Abroad Data Analysis Report/Summary

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Date: August 20th, 2025

This analysis explores trends in WSU Education Abroad from 2008 to 2025, examining participation over time, program types, academic outcomes, and destination patterns. After standardizing fields (e.g., year and major), engineering time bins, and addressing missing values where appropriate, I visualize the data through tables and charts with brief narratives. I then add a recent (2020–2025) snapshot and a policy-oriented equity view that compares regional access by college and major. The goal is not only to describe what happened, but to surface practical questions for planning, access, and program design.

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1. About Education Abroad (Context)

- Vision

We promote positive lifelong global awareness and engagement for the WSU community.

- Mission

We present the WSU community with the tools to explore and understand one's role in our global society. We commit to creating equitable opportunities for all identities to engage in global experiences. To this end, we provide a framework to develop global and international competencies, both on and off the WSU campuses. We believe this work is critical to the ongoing formation of a sustainable and peaceful world.

- Commitment to Diversity

We are committed to creating equitable opportunities for all identities within the WSU community to engage in global experiences.

source: https://ip.wsu.edu/education-abroad/

2. Quick Glossary

- Exchange: Students leaving WSU on bilateral exchange agreements
- Incoming: Students coming to WSU on bilateral exchange agreements
- Faculty-Led: Programs led by WSU faculty members and offering WSU credits
- Provider: Students traveling with a third-party provider
- NAP/ITR/Direct Enroll: Students traveling on one of our independent support options

3. Cleaning Data

3.1 Standardize column names

3.2 Build year and year bin from application cycle

- adjust fall terms to the following academic year (e.g., Fall 2016 -> Academic Year 2017)
- years into time periods (2008-2013, 2014-2019, 2020-2025)

3.3 Impute GPA using mean

- GPA global mean used for imputation: 3.2792038124578795
- Missing GPA before: 538, after: 0

3.4 Major name cleaning

a. Make the list: Save majors from the PDF as a simple .txt (one per line).
b. Clean the names: Lowercase everything and remove extra spaces.
c. Create a lookup: Map each cleaned name to the official major name (use first 4 letters as a quick helper, but double-check).
d. Match your data: Clean majors in your dataset the same way and replace them using the lookup; unmatched → "Other."
e. Fix leftovers: Print the unique items in "Other," add them to the lookup, and re-run.
f. Repeat quickly: Keep notes on what you changed so next time it's faster.

3.5 Check cleaning result (sample)

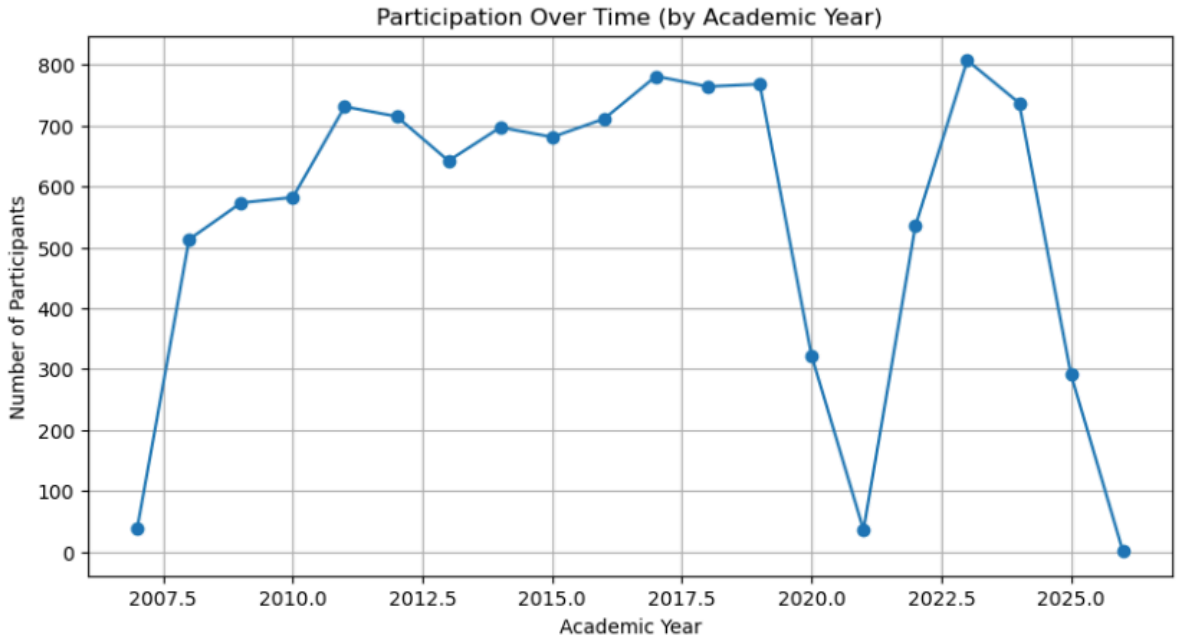
```
Columns in cleaned dataset:
['program', 'program_type', 'academic_year', 'app_cycle',
'preferred_pronouns', 'itinerary_start', 'location', 'country',
'student_cumulative_gpa', 'major', 'college', 'year', 'year_academic',
'year_bin']

Columns in df: ['Program', 'Program Type', 'Academic Year', 'App Cycle',
'Preferred Pronouns', 'Itinerary Start', 'Location', 'Country', 'Student
Cumulative GPA', 'Major', 'College', 'major_clean', 'major_std']
```

4. analyzing data

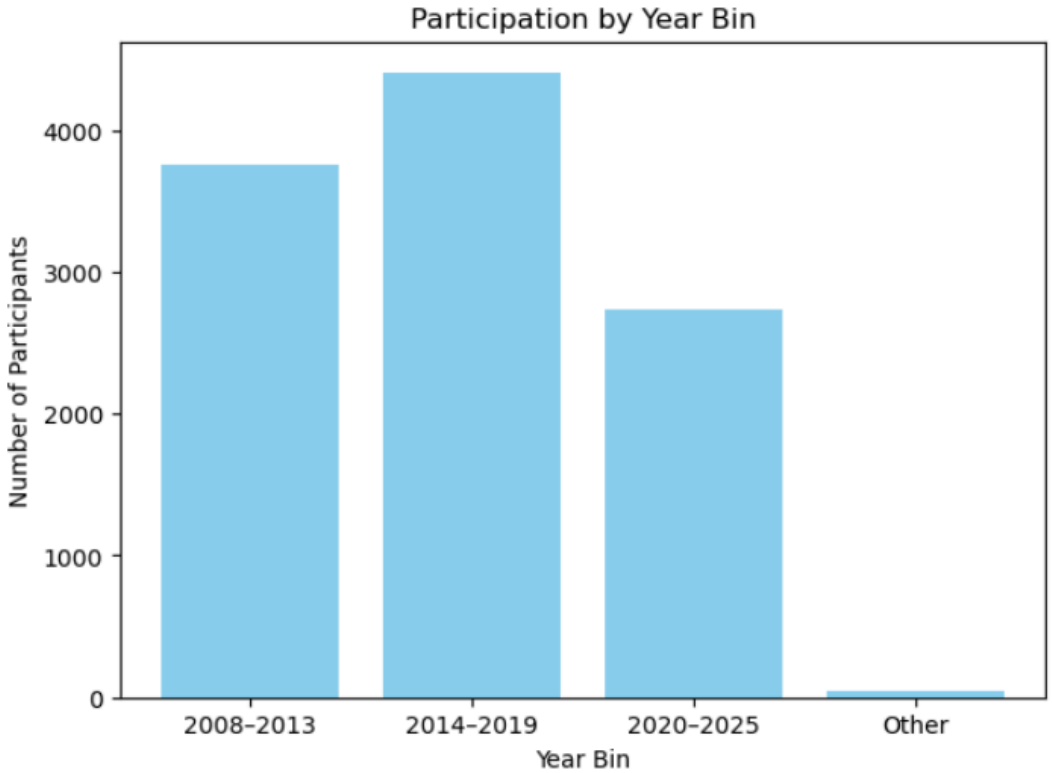
4.1 participation over time

```
Participation by Academic Year
   year_academic  count
0         2007.0     38
1         2008.0    513
2         2009.0    573
3         2010.0    582
4         2011.0    731
```



participation by year bin

```
Participation by Year Bin
    year_bin  count
0  2008-2013   3756
1  2014-2019   4402
2  2020-2025   2728
3      Other     39
```



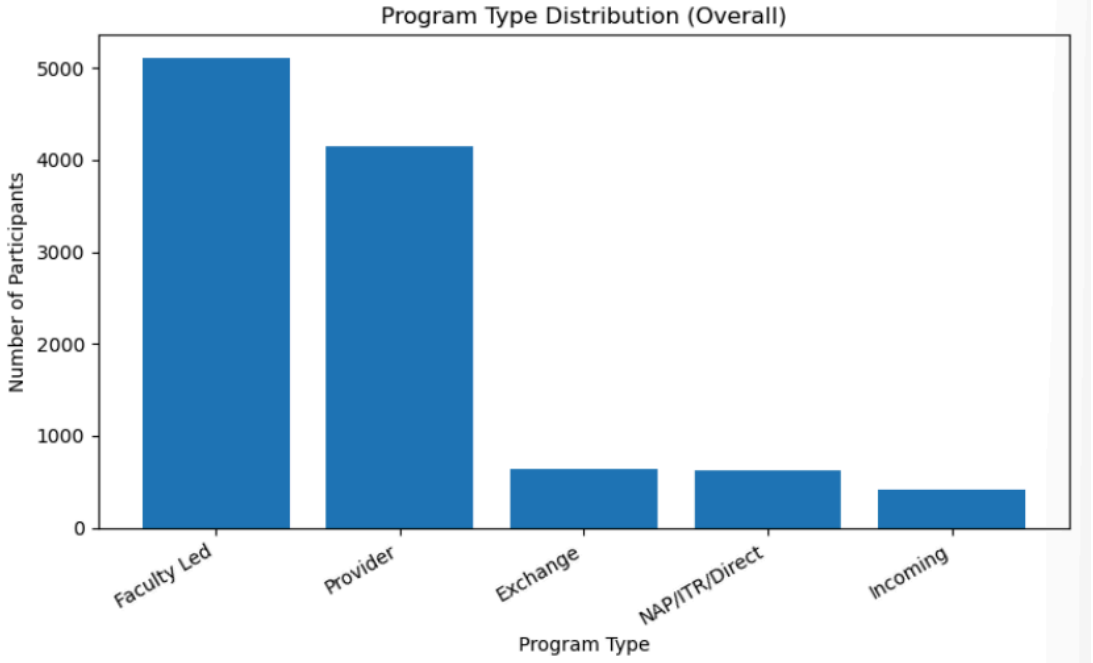
Insights

- Participation in study abroad programs shows steady growth from 2008 through the late 2010s, peaking around 2018–2019.
- The sharp decline around 2020–2021 aligns with the COVID-19 pandemic, which significantly disrupted international travel and academic exchanges.
- After 2022, participation began to recover, though the levels are not yet as high as the pre-pandemic peak.
- By year bins, 2014–2019 recorded the highest number of participants, followed by 2008–2013. The 2020–2025 period shows lower participation, largely due to pandemic disruptions, though recent years suggest signs of recovery.

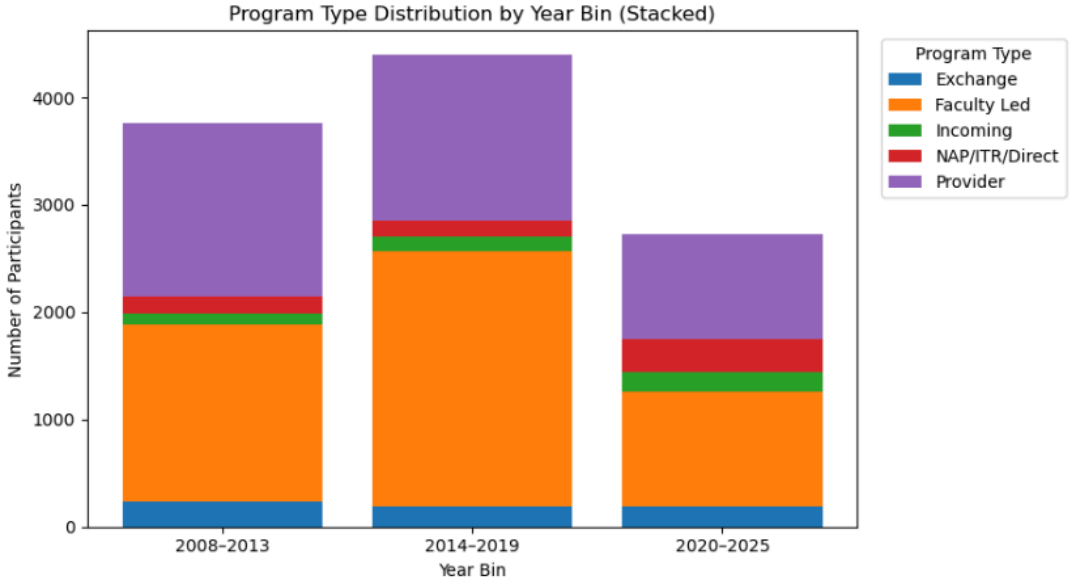
4.2 Participant characteristics

4.2.1 Program Type Distribution (overall + by year bin)

```
Program Type — Overall
     program_type  count
1     Faculty Led   5101
4        Provider   4149
0        Exchange    637
3  NAP/ITR/Direct    619
2        Incoming    419
```




```
Program Type by Year Bin (counts)
program_type  Exchange  Faculty Led  Incoming  NAP/ITR/Direct  Provider
year_bin
2008–2013          233         1653       100             163      1607
2014–2019          191         2377       138             141      1555
2020–2025          188         1071       181             313       975
```

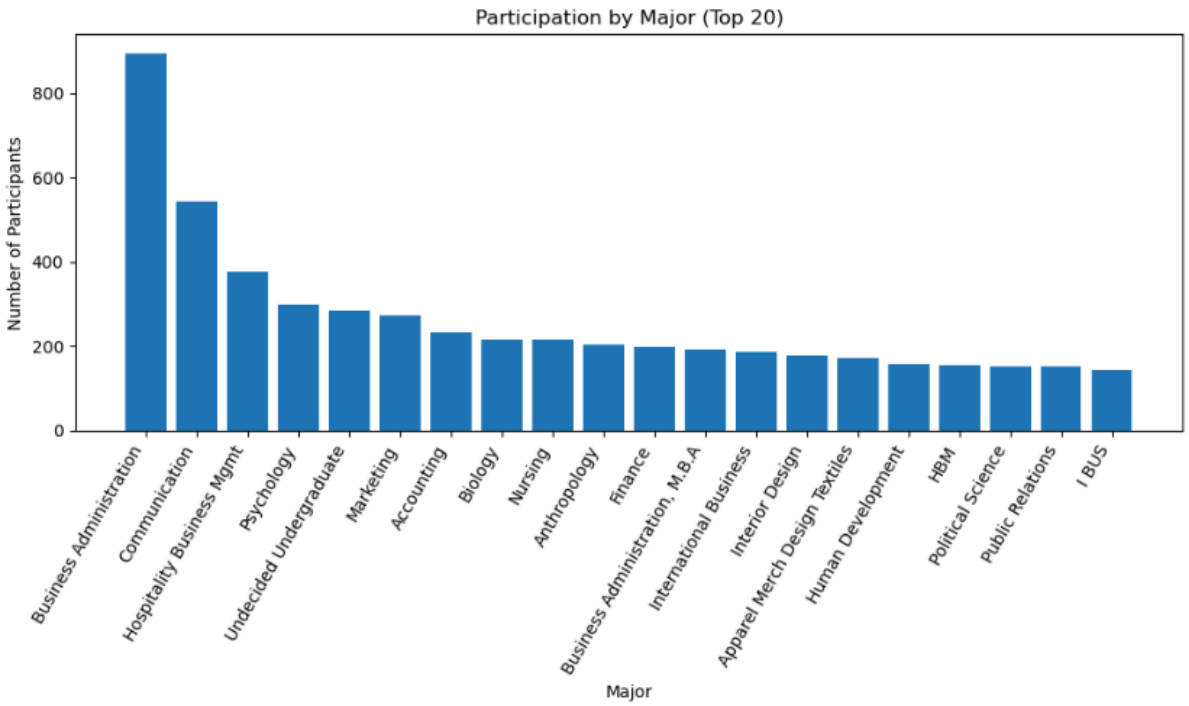


Insights

- Faculty-Led and Provider programs dominate participation, making up the majority across all time periods.
- Exchange, Incoming, and NAF/IF/Direct programs remain relatively small in scale compared to the two dominant types.
- The distribution by year bin shows that Provider programs were especially strong in earlier years (2008–2013), while Faculty-Led gained more prominence in recent periods.

4.2.2 Participation by Major

```
Participation by Major (using 'major') — Top 20
                                major  count
43            Business Administration    895
54                      Communication    544
144         Hospitality Business Mgmt    378
252                        Psychology    299
295           Undecided Undergraduate    284
182                         Marketing    273
5                        Accounting    232
39                          Biology    217
212                         Nursing    216
14                      Anthropology    204
106                         Finance    198
44    Business Administration, M.B.A    194
159           International Business    186
156                  Interior Design    177
15     Apparel Merch Design Textiles    174
146                Human Development    157
137                              HBM    155
244                Political Science    153
254                 Public Relations    151
148                              I BUS    143
```



Insight

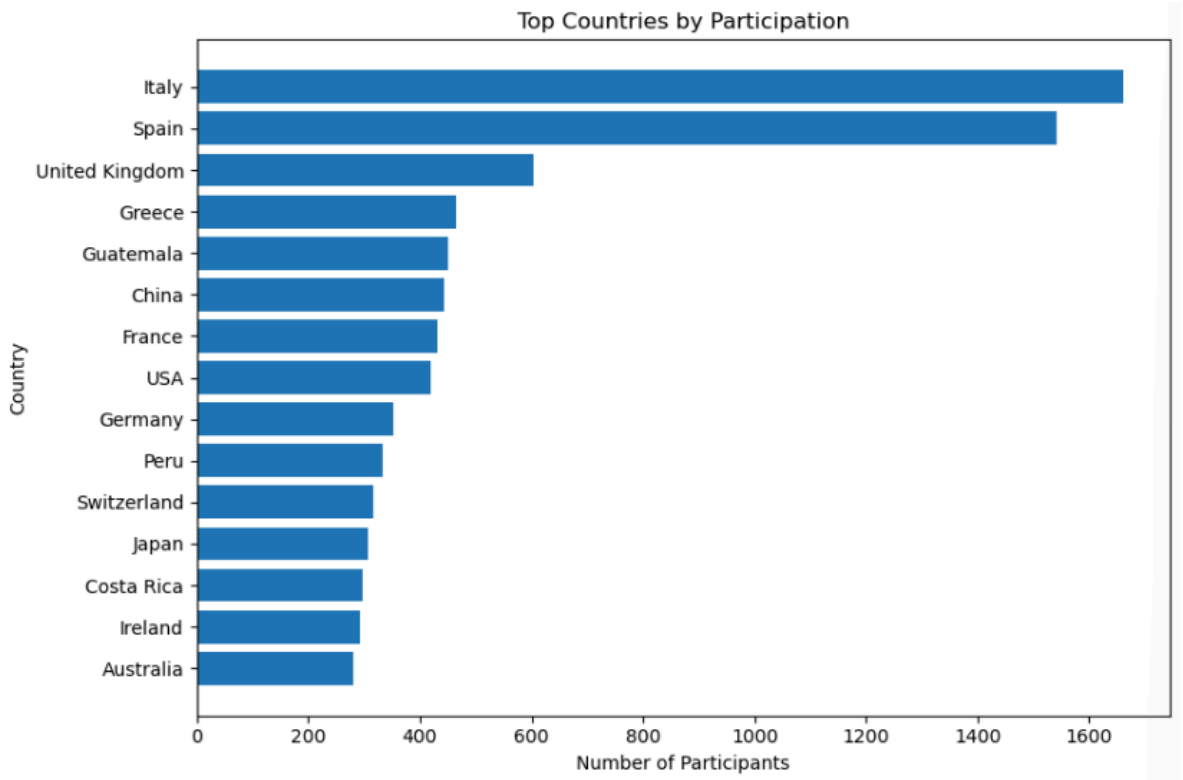
- Business Administration is by far the largest major represented, followed by Communication and Hospitality/Business tracks.

- Psychology, Nursing, and Economics are also strong contributors, reflecting both social sciences and professional programs.
- A “long tail” of majors exists, where participation is spread across diverse academic disciplines, but at smaller scales.

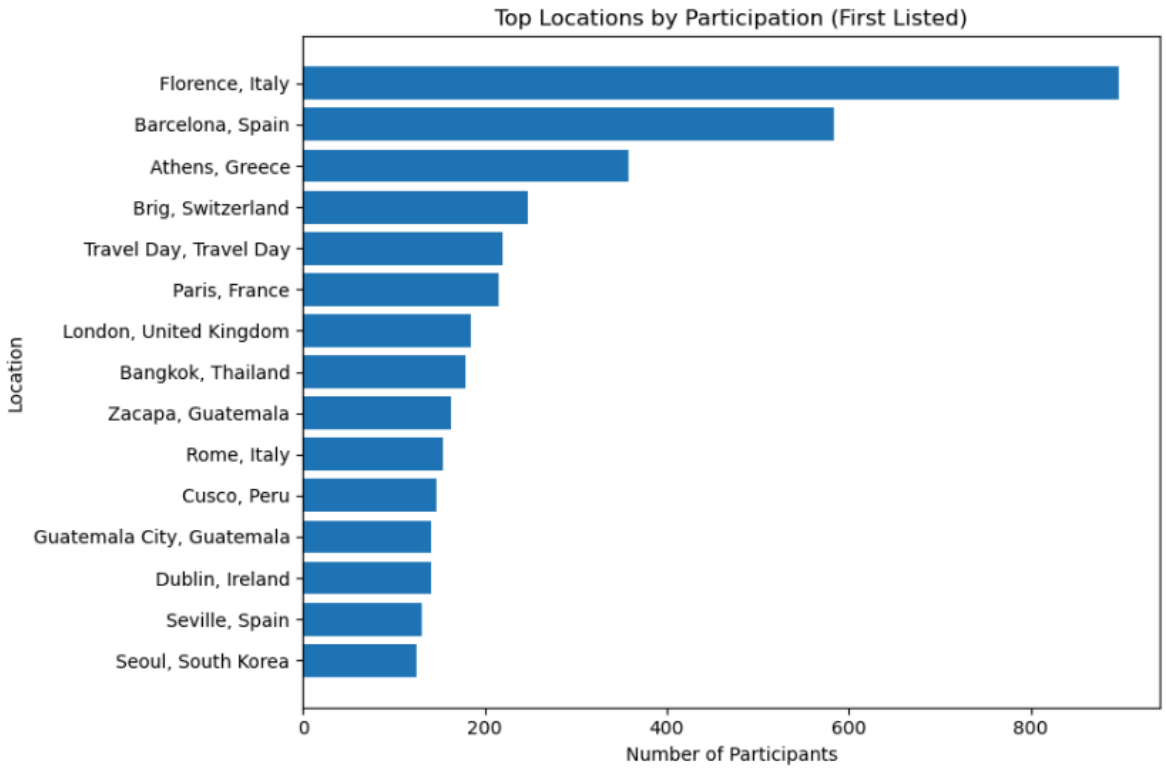
4.3 Geographic distribution

4.3.1 Top Locations & Countries

```
Top Countries (by participants)
            country  count
59            Italy   1662
99            Spain   1542
114  United Kingdom    603
47           Greece    464
49        Guatemala    450
22            China    444
41           France    430
111             USA    419
43          Germany    353
86             Peru    334
103     Switzerland    315
61            Japan    307
25       Costa Rica    297
57          Ireland    292
6         Australia    281
```



```
Top Locations (first listed city)
                      location  count
120            Florence, Italy    897
34            Barcelona, Spain    584
23              Athens, Greece    358
55           Brig, Switzerland    247
362     Travel Day, Travel Day    219
266              Paris, France    215
206     London, United Kingdom    185
32           Bangkok, Thailand    178
404          Zacapa, Guatemala    162
295                Rome, Italy    154
95                 Cusco, Peru    146
144  Guatemala City, Guatemala    141
108            Dublin, Ireland    141
321             Seville, Spain    131
319         Seoul, South Korea    125
```



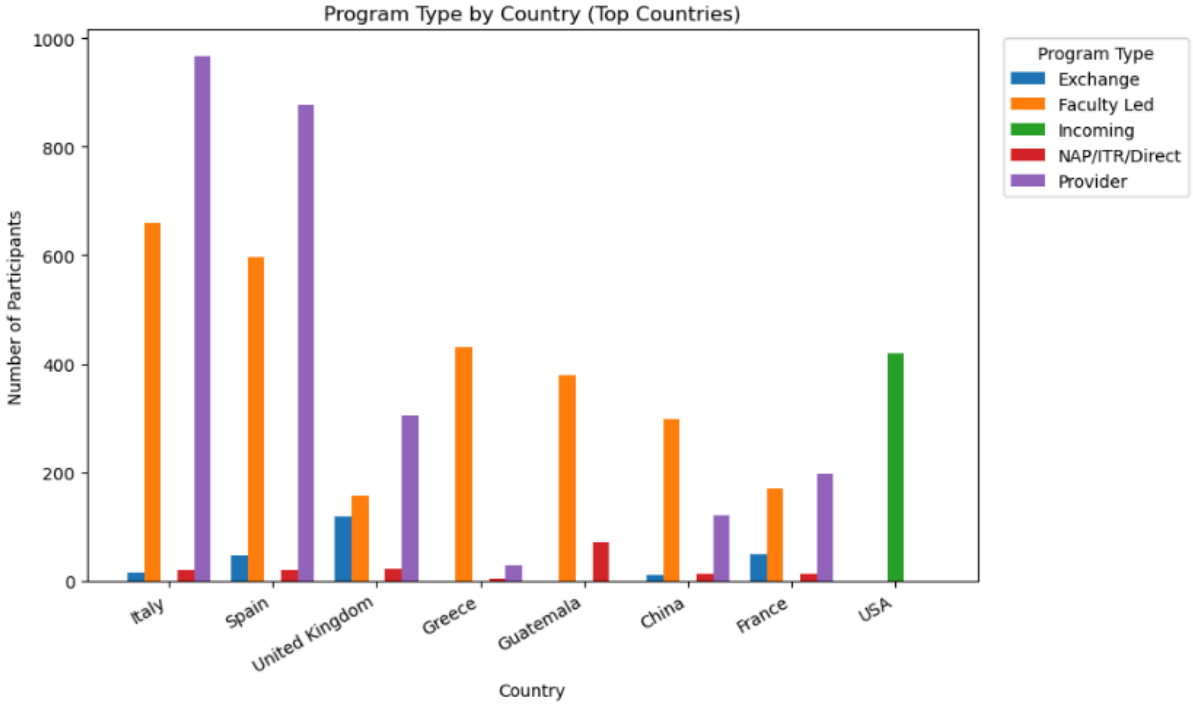
Insights

- Countries: Italy and Spain dominate participation, far ahead of others (both ~1,500+ students). The UK, Greece, and Guatemala form a second tier, while destinations like Japan, Costa Rica, and Australia attract smaller groups.

- Cities: Florence (Italy) and Barcelona (Spain) are clear hubs, with Athens and Switzerland's Brig also standing out. European capitals (Paris, London, Rome) remain consistently popular.
- There is a strong European bias, especially toward Mediterranean destinations. A small number of cities attract the bulk of students, which may indicate concentration in well-established partner institutions/programs.

4.3.2 Program Type x Country

```
Program Type × Country (Top Countries)
program_type    Exchange  Faculty Led  Incoming  NAP/ITR/Direct  Provider
country
Italy                15          660         0              20       967
Spain                47          597         0              20       878
United Kingdom      118          157         0              23       305
Greece                0          431         0               4        29
Guatemala             0          379         0              71         0
China                11          299         0              13       121
France               50          170         0              12       198
USA                   0            0       419               0         0
```



Insights

- Italy and Spain show high participation through Provider and Faculty-led programs.
- The UK has more Exchange participants than most other countries.
- Guatemala is mainly associated with Faculty-led programs, while the USA is unique for Incoming students.

- Program type distribution is destination-specific. Europe leans heavily on provider-led/faculty-led structures, while countries like the USA reflect inbound participation patterns.

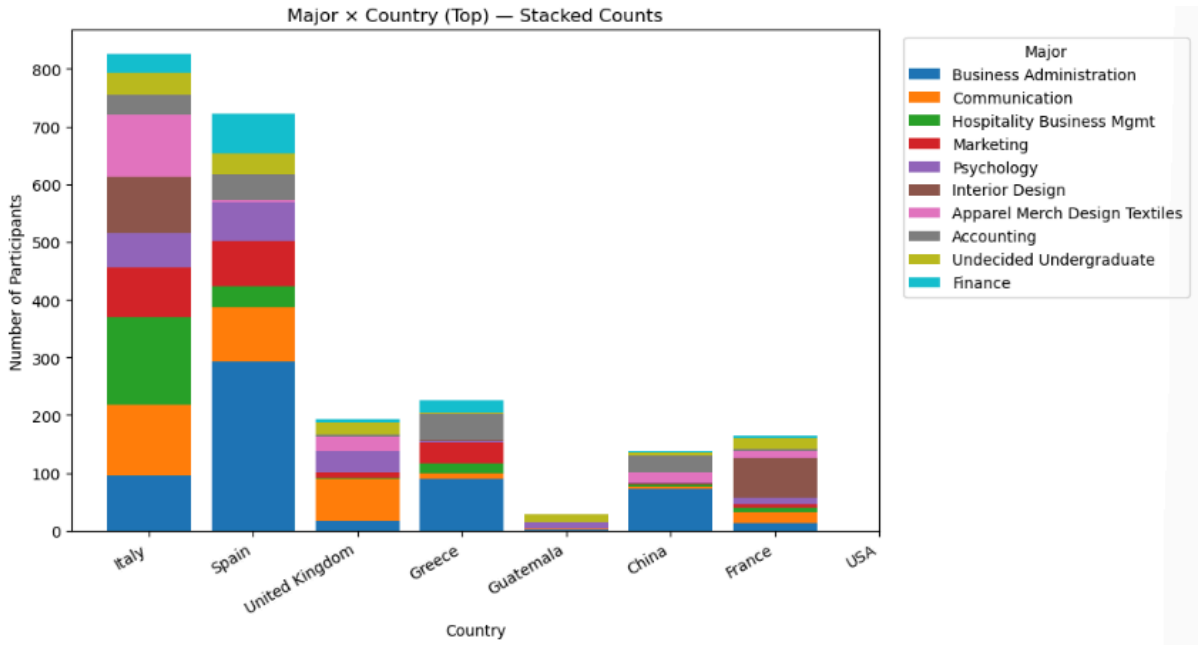
4.3.3 Major x Country

```
Major × Country (Top countries × Top majors) — counts
major           Business Administration  Communication  \
country
Italy                              96.0          122.0
Spain                             292.0           94.0
United Kingdom                     16.0           74.0
Greece                             90.0            9.0
Guatemala                           1.0            3.0
China                              72.0            5.0
France                             13.0           19.0
USA                                 NaN            NaN

major           Hospitality Business Mgmt  Marketing  Psychology  \
country
Italy                              152.0       85.0        60.0
Spain                               38.0       77.0        68.0
United Kingdom                       2.0        9.0        36.0
Greece                              17.0       37.0         2.0
Guatemala                            0.0        0.0        11.0
China                                4.0        1.0         2.0
France                               7.0        7.0        12.0
USA                                  NaN        NaN         NaN

major           Interior Design  Apparel Merch Design Textiles  Accounting  \
country
Italy                      98.0                          108.0        33.0
Spain                       0.0                            3.0        44.0
United Kingdom              1.0                           24.0         5.0
Greece                      1.0                            0.0        46.0
Guatemala                   0.0                            0.0         0.0
China                       0.0                           17.0        29.0
France                     68.0                           11.0         4.0
USA                         NaN                            NaN         NaN

major           Undecided Undergraduate  Finance
country
Italy                              39.0     33.0
Spain                              38.0     68.0
United Kingdom                     21.0      5.0
Greece                              3.0     21.0
Guatemala                          14.0      0.0
China                               5.0      3.0
France                             19.0      5.0
USA                                 NaN      NaN
```



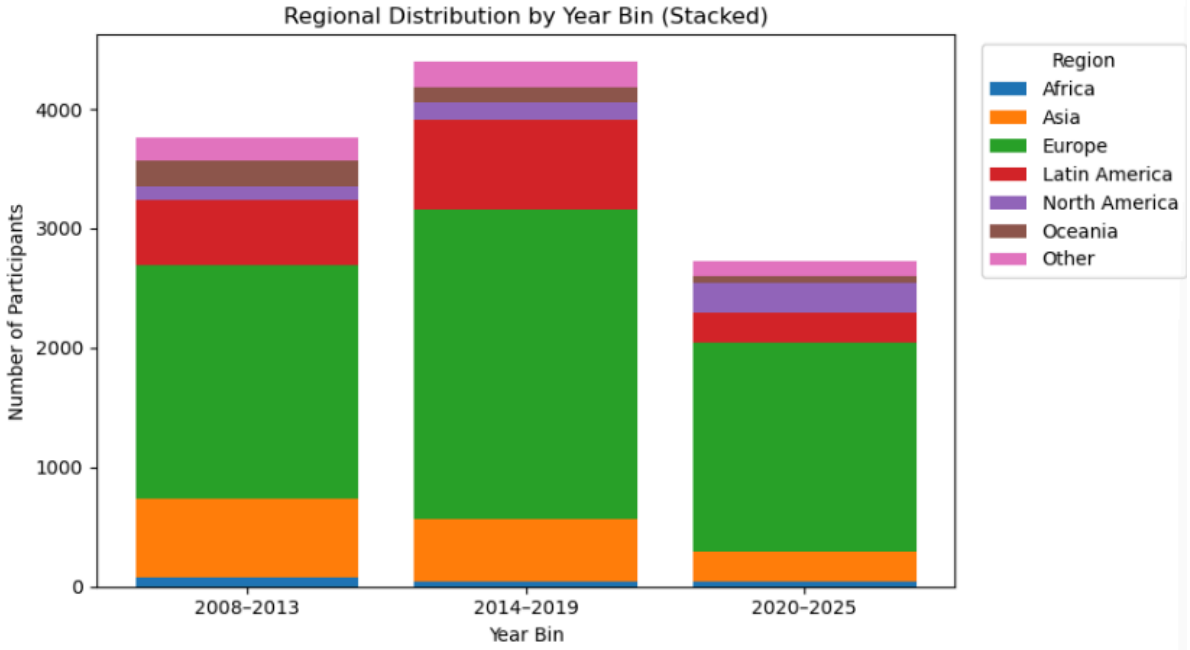
Insights

- Italy and Spain dominate participation volumes, with Italy slightly ahead. These two destinations account for the bulk of study abroad activity.
- Business Administration is consistently strong across most countries, but especially in Spain and Greece.
- Hospitality Business Management stands out in Italy, making Italy the most popular destination for that major.
- Interior Design and Apparel/Textiles have a large share in Italy, suggesting strong alignment with Italy's cultural/academic reputation in design and fashion.
- Spain shows more variety in majors compared with Italy while Business still leads, Communication, Psychology, and Finance are also significant contributors.
- Other countries (UK, Greece, Guatemala, China, France) show smaller participation but still reveal distinct patterns. e.g., UK with Communication, Greece with Business/Marketing, France with Interior Design.
- Italy = design/fashion + hospitality hub, Spain = business + broad mix, while other destinations remain niche but with recognizable academic alignments.

4.3.4 Regional Bias (continent-level aggregation

```
Region distribution (overall)
          region  count
0         Europe   6320
1  Latin America   1561
2           Asia   1438
3          Other    544
4  North America    501
5        Oceania    395
6         Africa    166

Region × Year Bin (counts)
region     Africa  Asia  Europe  Latin America  North America  Oceania  Other
year_bin
2008-2013      73   666    1949            555            114      209    190
2014-2019      48   513    2594            758            142      122    225
2020-2025      45   252    1751            248            245       61    126
```

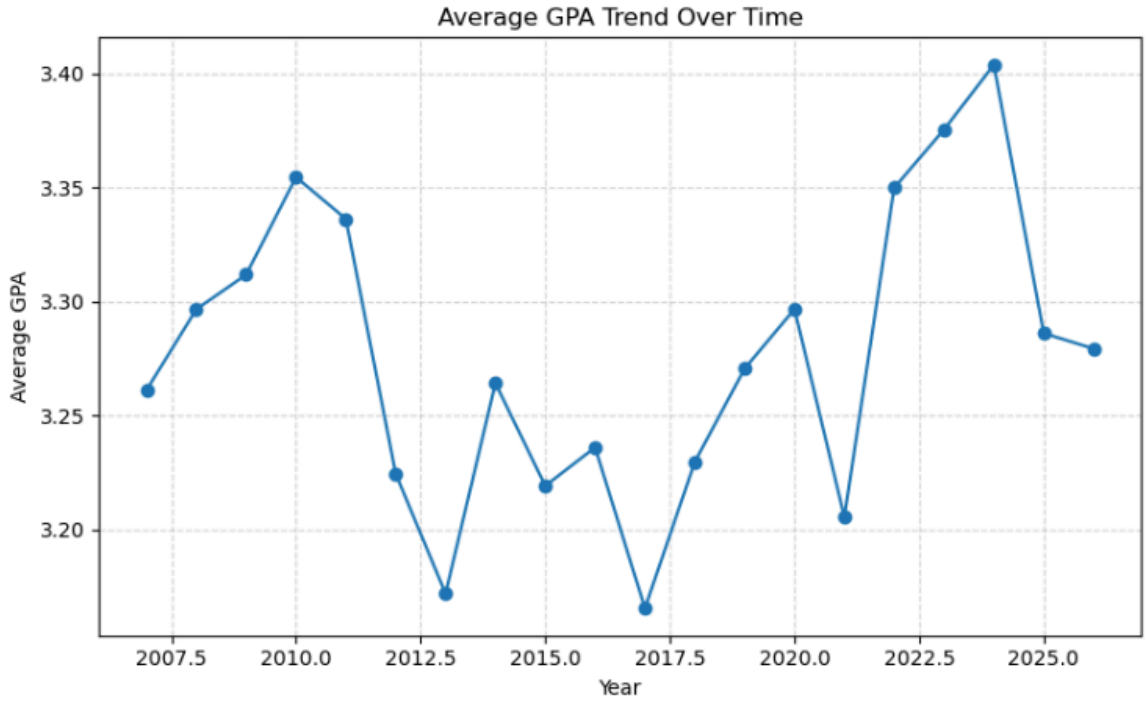


Insights

- Europe dominates across all year bins, accounting for the majority of participants.
- Latin America and Asia consistently attract smaller but steady numbers.
- Africa and Oceania remain marginal destinations.
- Participation drops after 2020 in all regions, consistent with COVID-era travel disruptions.
- Study abroad participation is heavily Eurocentric, with limited regional diversity. Although Asia and Latin America are present, they remain secondary destinations, and Africa is very underrepresented.

4.4 Academic & Performance

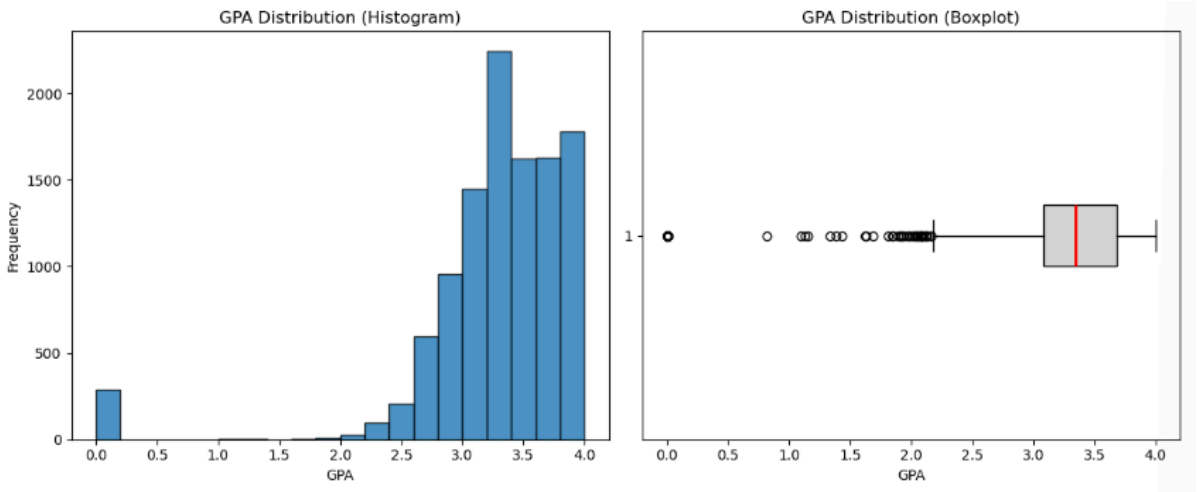
4.4.1 GPA Trend Over Time



Insights

- The average GPA has remained relatively stable between 3.2–3.35 over the years.
- There is noticeable volatility: some dips around 2013–2015, followed by recovery and a peak around 2021–2022 (close to 3.4).
- The more recent decline (2023 onward) might suggest academic or programmatic challenges post-pandemic.

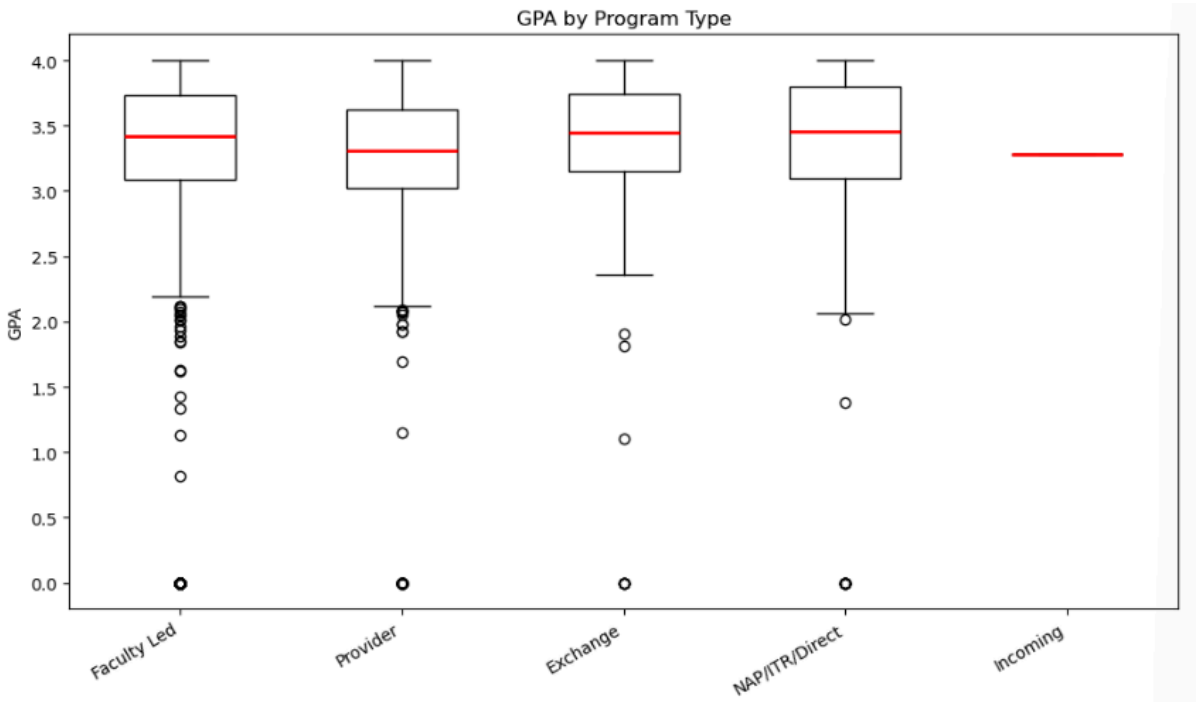
4.4.2 GPA Distribution



Insights

- Most students cluster around the 3.0–3.7 GPA range, indicating strong overall academic performance.
- Very few students fall below 2.0, showing that low academic performance is rare.
- The boxplot confirms a tight distribution around 3.3, with some outliers on both low and high ends.

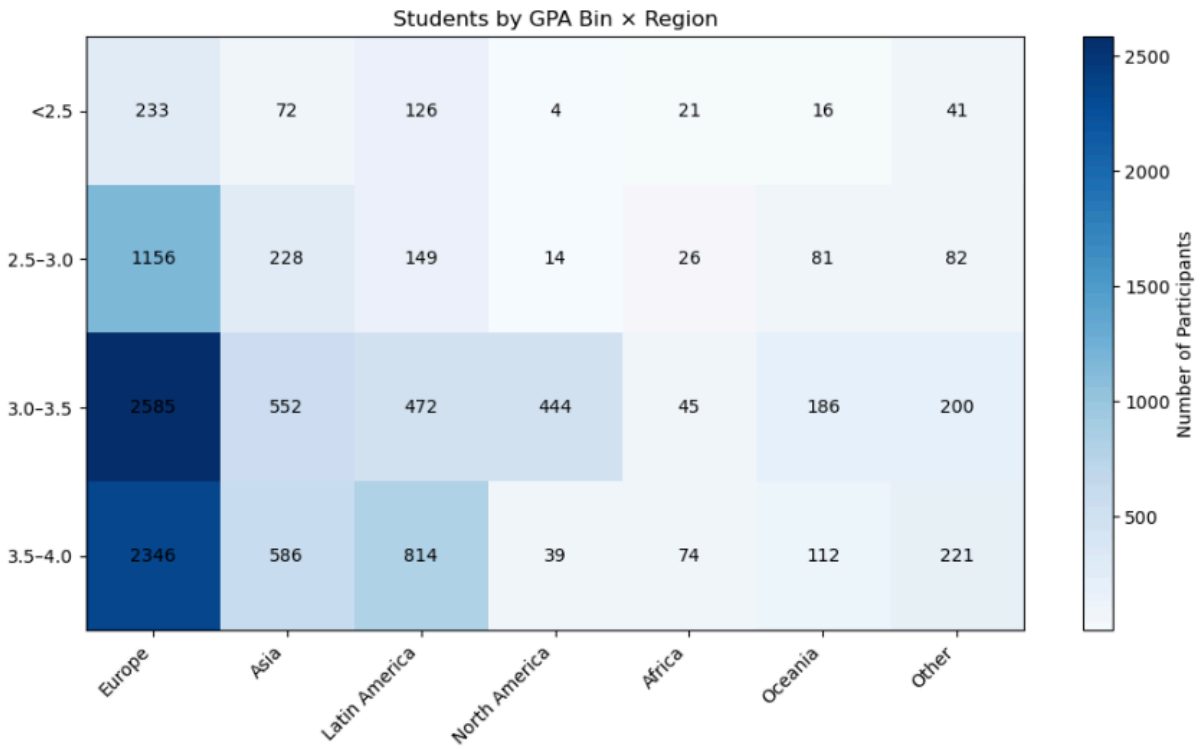
4.4.3 GPA by Program Type



Insights

- Faculty-led and Provider programs show slightly higher median GPAs compared to Exchange and Internship programs.
- Internship participants are few in number but exhibit a flat boxplot, meaning their GPAs are clustered without much variation.
- Exchange programs show more variability, with a wider spread and more low-GPA outliers—suggesting that exchange students face more academic challenges or grading differences abroad.

4.4.4 GPA bin x Region



Insights

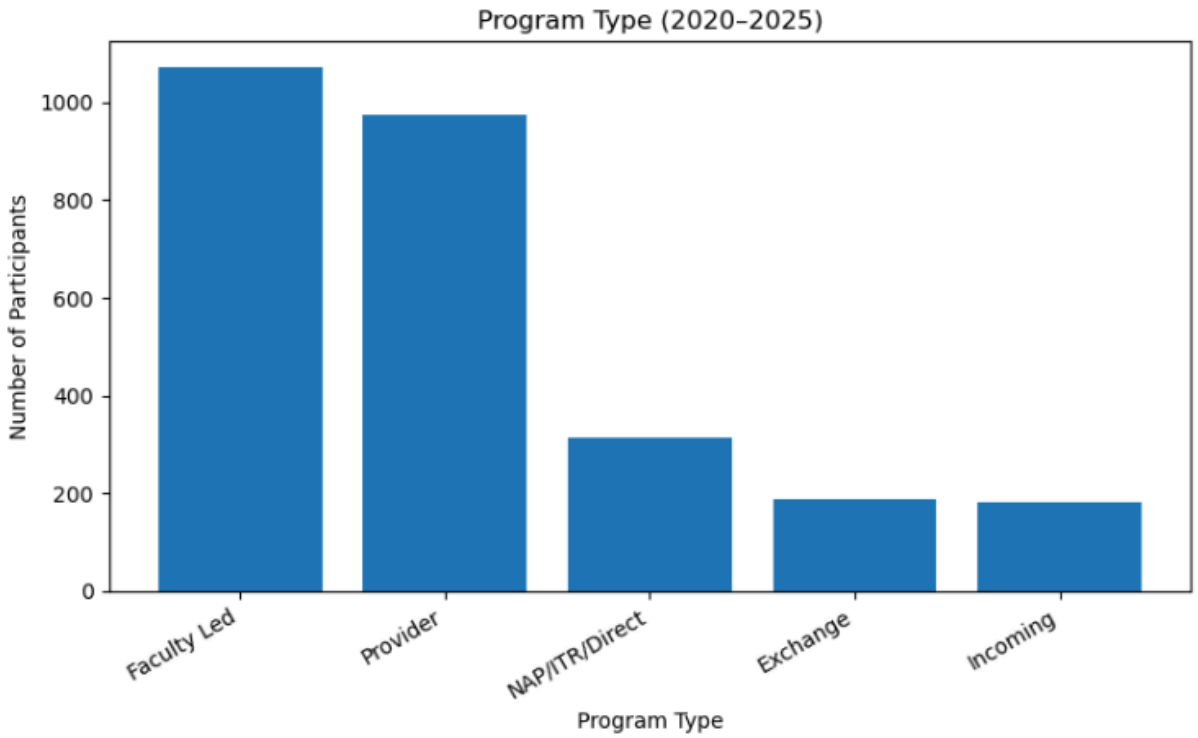
- Europe dominates across all GPA bins, but especially in the 3.0–3.5 and 3.5–4.0 ranges, showing that the majority of students with higher GPAs are concentrated in European programs.
- Asia also shows strong representation across GPA bins, but with a relatively higher share in the 2.5–3.0 range compared to Europe.
- Latin America has a balanced spread, with notable presence in both the 3.0–3.5 and 3.5–4.0 bins.
- North America programs are small in volume (likely fewer abroad programs in the same region) but concentrated in mid-to-high GPAs.

- Oceania (Australia, New Zealand, Fiji) also shows a good number of students in the 3.5–4.0 bin, suggesting strong student outcomes.
- Africa has fewer participants overall, but those who join are fairly evenly spread, including high GPAs.

4.5 Recent Snapshot (2020-2025)

4.5.1 Program Type

```
Recent Snapshot (2020-2025) — Program Type counts
     program_type  count
1     Faculty Led   1071
4        Provider    975
3  NAP/ITR/Direct    313
0        Exchange    188
2        Incoming    181
```

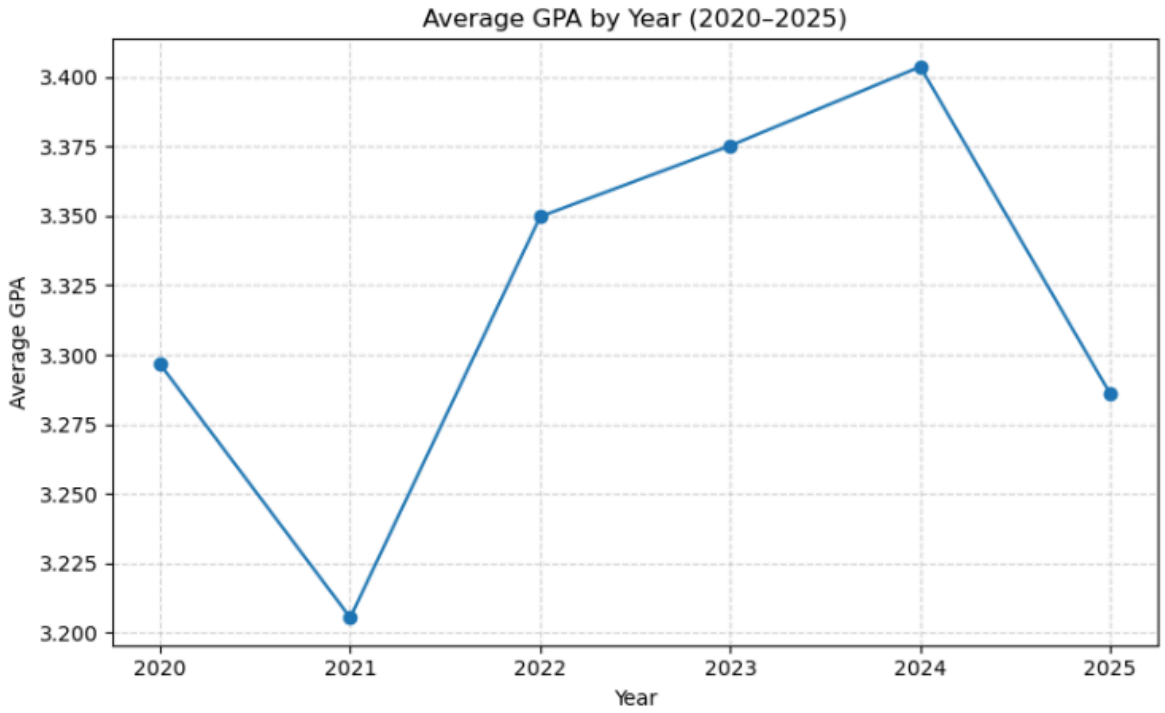


Insights

- Faculty-Led (1,071; 39%) and Provider (975; ~36%) dominate the post-2020 mix; NAP/ITR/Direct (12%) is a clear mid-tier; Exchange (7%) and Incoming (7%) are small.
- This composition suggests recovery after COVID was anchored by shorter, more structured models (faculty-led/provider) rather than traditional exchanges.

4.5.2 Average GPA by Year (2020-2025)

```
Recent Snapshot (2020-2025) — Average GPA by Year
   year_academic  student_cumulative_gpa
0         2020.0                3.296546
1         2021.0                3.205423
2         2022.0                3.349758
3         2023.0                3.375286
4         2024.0                3.403701
5         2025.0                3.285941
```



Insights

- GPA dips in 2021 (3.21), then rebounds steadily to a peak in 2024 (3.40) before a modest pullback in 2025 (~3.29).
- Overall variance is narrow (~3.21–3.40), indicating stable academic performance among recent participants despite pandemic disruptions.

4.5.3 Quick participation summary

```
Recent Snapshot (2020-2025) — Participation counts by year
   year_academic  count
0         2020.0    322
1         2021.0     36
2         2022.0    535
3         2023.0    807
4         2024.0    737
5         2025.0    291
```

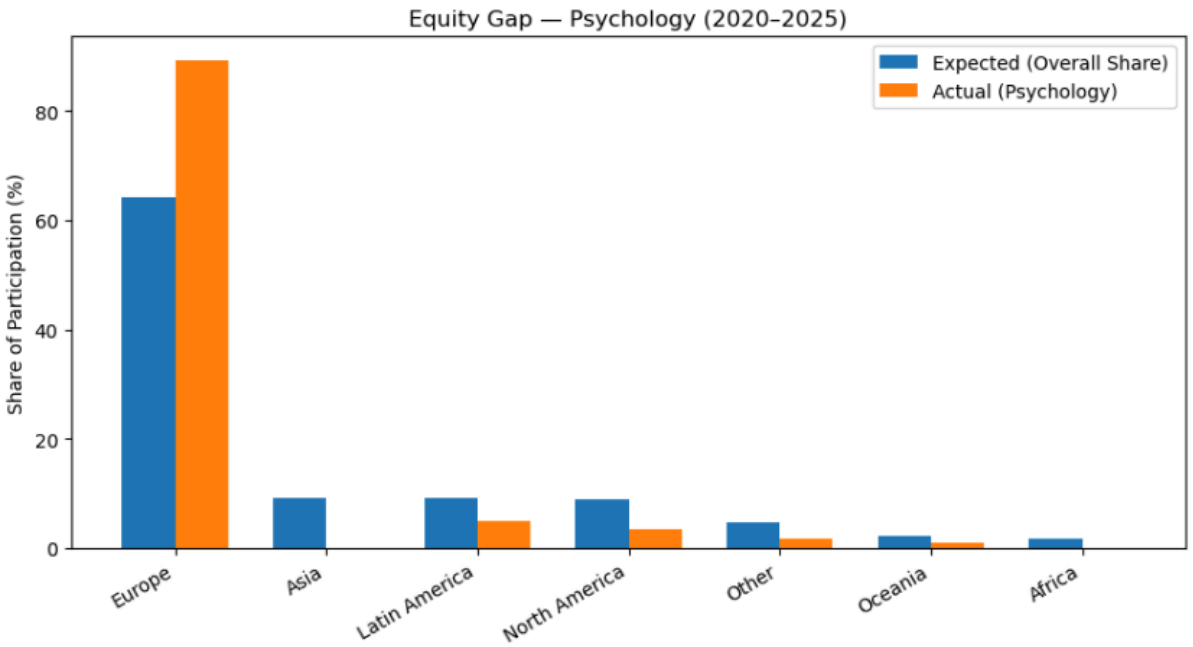
Insights

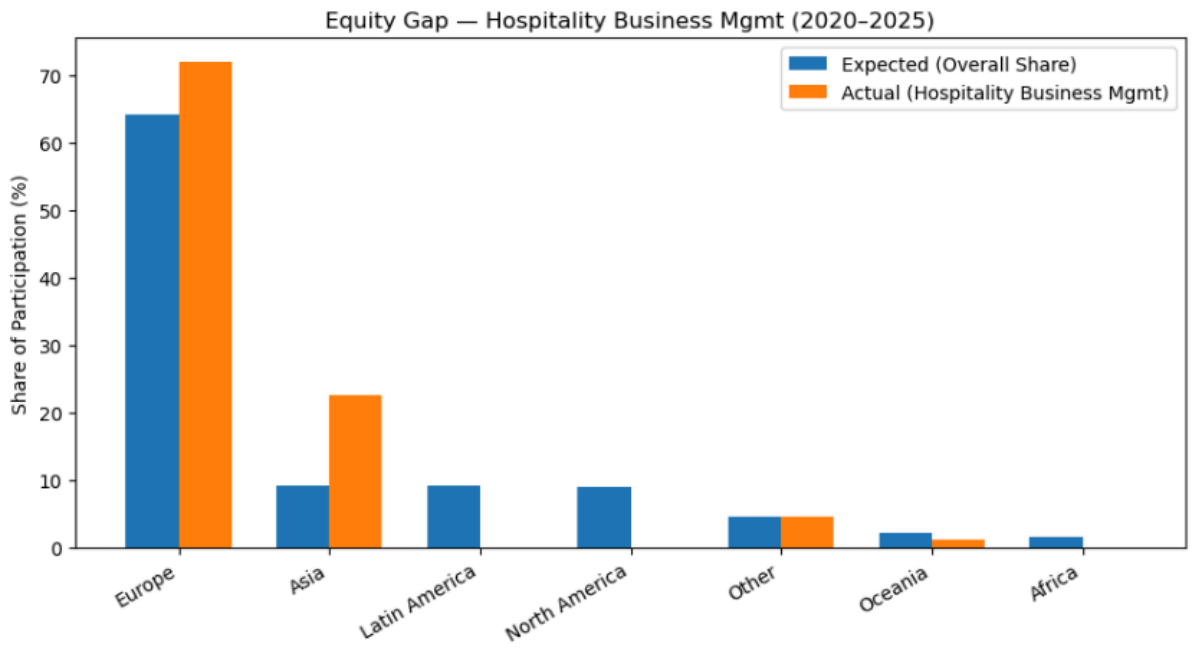
- 2021 collapses (36 students)—pandemic impact. Strong rebound in 2022 (535) and 2023 (807); 2024 remains high (737).
- 2025 is lower (291), likely a partial year in the dataset; watch year-end totals to confirm trend.

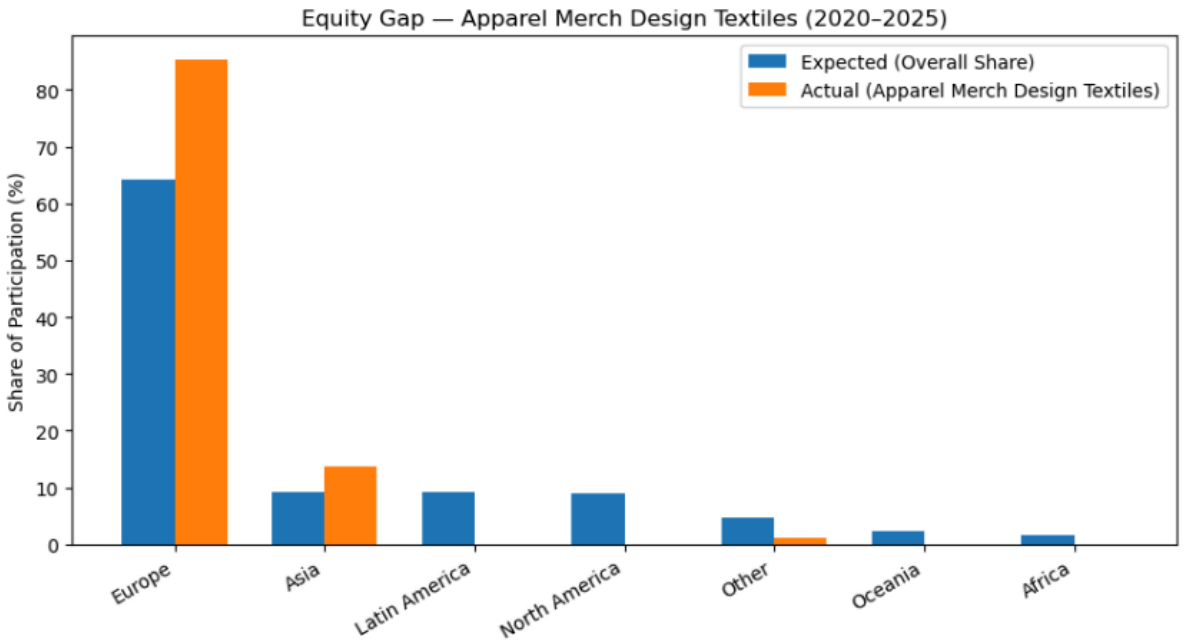
4.6 Policy-Oriented Analysis (Equity Gaps & Suggestions)

```
Overall region composition (2020–2025) — expected shares
region
Europe           64.2%
Asia              9.2%
Latin America     9.1%
North America     9.0%
Other             4.6%
Oceania           2.2%
Africa            1.6%
Name: overall_share, dtype: object

major × Region — largest negative gaps (under-representation)
                            major         region  gap_pp  count  total
72                        Unknown         Europe   -61.3      6    206
16                        Biology         Europe   -25.7     40    104
57                     Psychology           Asia    -9.2      0    120
31      Hospitality Business Mgmt  Latin America    -9.1      0     89
10  Apparel Merch Design Textiles  Latin America    -9.1      0     88
66               Public Relations  Latin America    -9.1      0     96
38         International Business  Latin America    -9.1      0    106
3                      Accounting  Latin America    -9.1      0     80
32      Hospitality Business Mgmt  North America    -9.0      0     89
4                      Accounting  North America    -9.0      0     80
11  Apparel Merch Design Textiles  North America    -9.0      0     88
53                       Marketing  North America    -9.0      0    193
67               Public Relations  North America    -9.0      0     96
15                        Biology           Asia    -8.3      1    104
52                       Marketing  Latin America    -8.1      2    193
```


Equity Gap — Psychology (2020–2025)
Expected (Overall Share)
Actual (Psychology)
Share of Participation (%)
0
20
40
60
80
Europe
Asia
Latin America
North America
Other
Oceania
Africa


Equity Gap — Hospitality Business Mgmt (2020–2025)
Expected (Overall Share)
Actual (Hospitality Business Mgmt)
Share of Participation (%)
0
10
20
30
40
50
60
70
Europe
Asia
Latin America
North America
Other
Oceania
Africa


Equity Gap — Apparel Merch Design Textiles (2020–2025)
Expected (Overall Share)
Actual (Apparel Merch Design Textiles)
Share of Participation (%)
0
10
20
30
40
50
60
70
80
Europe
Asia
Latin America
North America
Other
Oceania
Africa

```
Regional diversification (HHI) by period (lower = more diversified)
year_bin
2008-2013    0.329490
2014-2019    0.395020
2020-2025    0.439756
Other        0.488494
dtype: float64
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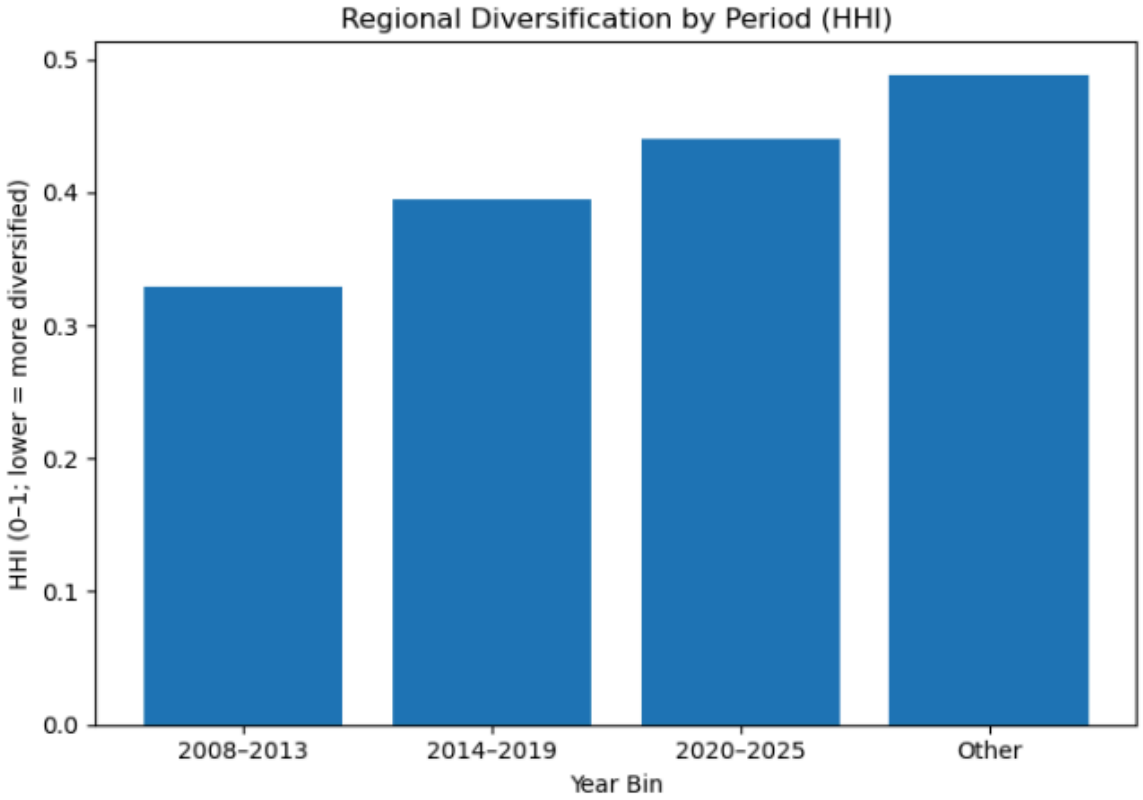

a. Interpreting the Results

- From 2020–2025, Europe accounts for 64% of all study abroad destinations, while Asia, Latin America, and North America each make up only around 9–10%.
- By major:
  - Psychology students study abroad in Asia far less than expected.
  - Hospitality Business Management and Apparel Design/Textiles students study abroad in Latin America and North America less often.
  - Other majors (Marketing, Biology, Accounting) also send relatively few students outside of Europe.
- The charts confirm that for many majors, study abroad opportunities are strongly concentrated in Europe, while destinations in Asia, Latin America, and Africa are under-used.

b. Explaining the Over-Representation of Europe

- Program structures and partnerships: WSU likely has more established exchange agreements, partner institutions, or logistical support in Europe, making it the most accessible option.
- Perceived safety and familiarity: Students and families may see Europe as a “safer” or “easier” destination compared to other regions.
- Major-specific appeal: Programs in Hospitality, Design, or Psychology may be particularly strong or attractive in Europe, reinforcing the imbalance.
- Institutional history: Long-standing European connections and alumni networks may make it harder for newer destinations to grow.

c. Diversity Over Time

The HHI index (measuring concentration of destinations) shows a clear trend:

- 2008–2013: more balanced use of regions.
- 2014–2019 and 2020–2025: increasing concentration in Europe, meaning fewer students explore beyond that region.

Instead of becoming more globally diverse, WSU’s study abroad programs have become more Eurocentric.

Possible reasons:

- Program growth focused mainly on Europe, not equally in Asia, Africa, or Latin America.
- Recruitment and marketing messages that highlight Europe more strongly.
- Barriers such as language, cost, or visa complexity make non-European regions less accessible.

d. Why promoting diversity policies is necessary

The mission of Education Abroad is to promote global awareness and engagement and to create equitable opportunities for all identities. If most students only go to Europe, they miss the chance to engage with the wider world(cultures, economies, and perspectives) in Asia, Africa, and Latin America.

Promoting diversity in destinations is therefore essential because:

- It ensures students gain a truly global perspective, not just a European one.
- It provides equitable access to under-represented regions, in line with the department’s commitment to diversity.
- It helps prepare WSU students for a world where future careers and collaborations will increasingly depend on connections beyond Europe, especially in Asia, Latin America, and Africa.

Without deliberate policies, the imbalance will continue, and the vision of preparing students for a sustainable and peaceful world will be harder to achieve.

e. Policy Suggestions

- Recruitment & scholarships: Target support for programs in Asia, Latin America, and Africa.
- Expand access beyond Europe: Build more partnerships and exchanges outside Europe.
- Make programs more global: Market the unique academic and cultural value of non-European destinations.
- Track progress: Monitor destination diversity by major and set goals to reduce over-reliance on Europe.

5. Further Analysis Ideas

- Ambassador Diversity Audit: Compare ambassador demographics and academic backgrounds to the participant pool. If certain groups are missing, recruit targeted ambassadors to bridge representation gaps.
- Scholarship Impact Study: Merge with scholarship award data to see if financial aid correlates with higher participation among historically under-represented groups and use this to inform funding priorities.
- Program Outcome Tracking: Follow up with participants on skills, career paths, or further study; identify which program types yield the strongest long-term benefits. (this could be done with open-ended questions survey)
- Geographic Risk & Opportunity Analysis: Combine destination data with safety indexes and partner availability to diversify options while managing risk.
- Demand Prediction: Use past trends by year, college, and program type to predict future demand and adjust program capacity and marketing strategies accordingly.
- Scholarship & Financial Aid Effects: Merge award data to test whether aid increases participation from low-income or underrepresented groups. Estimate effect sizes by college/major and year bins to guide funding allocations and outreach timing.
- Add campus enrollment denominators by college/major to compute true participation rates.